# Does change in ethical education influence core moral values? Towards culture-aware morality model

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**Abstract**

In this study, we focus on ethical education as a means to improve artificial companion's conceptualization of moral decision-making process in human users. In particular, we focus on automatically determining whether changes in ethical education influenced core moral values in humans throughout the century. We analyze ethics as taught in Japan before WWII and today to verify how much the pre-WWII moral attitudes have in common with those of contemporary Japanese, to what degree what is taught as ethics in school overlaps with the general population's understanding of ethics, as well as to verify whether a major reform of the guidelines for teaching the school subject of "ethics" at school after 1946 has changed the way common people approach core moral questions (such as those concerning the sacredness of human life). We selected textbooks used in teaching ethics at school from between 1935 and 1937, and those used in junior high schools today (2019) and analyzed what emotional and moral associations such contents generated. The analysis was performed with an automatic moral and emotional reasoning agent and based on the largest available text corpus in Japanese as well as on the resources of a Japanese digital library. As a result, we found out that, despite changes in stereotypical view on Japan's moral sentiments, especially due to historical events, past and contemporary Japanese share a similar moral evaluation of certain basic moral concepts, although there is a large discrepancy between how they perceive some actions to be beneficial to the society as a whole while at the same time being inconclusive when it comes to assessing the same action's outcome on the individual performing them and in terms of emotional consequences. Some ethical categories, assessed positively before the war while being associated with a nationalistic trend in education have also disappeared from the scope of interest of post- war society. The findings of this study support suggestions proposed by others that the development of personal AI systems requires supplementation with moral reasoning.

**Keywords:** machine ethics, moral education, automatic moral reasoning

## 1 Introduction

The concept of *machine ethics* is fairly young[1] and as such, discussion concerning ethical issues surrounding Artificial Intelligence (AI) is still at an early stage, especially when compared to the amount of intellectual debate and financial investment being made in the technological side of AI. Nevertheless, recent rapid developments in the field of AI surfaced the issues previously considered to arise much later

[1] The issues surrounding ethics in AI were first approached by [Rzepka and Araki, 2005] and later by [Anderson *et al*., 2006].

in the future and created the need to redefine what would constitute a moral and ethical common ground in the further research surrounding AI [LaChat, 1986].

Since moral and ethical education is an indispensable element in the formation of the character of any human being, any AI system whose task is to accompany humans by definition is obliged to have an ethical insight as well. However, reducing complex ethical and moral questions to a set of universal rules or core moral values poses a big challenge to humans, let alone today's AI systems.

One might also wonder why we would find the idea of providing artificial companions with deep knowledge concerning shifts in ethics as understood by the broader audience important, given that the question of implementation of spirituality or understanding of religious experience in machines today seems still far-fetched. However, exploring today how technology is predictable to develop in the areas of ethics and understanding of morality, and what influences people's moral choices and preferences, is important for appropriate appraisal and appreciation prior to having the developments already materialize, and then scrambling to react. This goes along with the especially recently noticeable new rising trend in the development of technology, namely to create the technology personalized so it was capable to respond to the private needs of a diversified range of users. Therefore, the designers and programmers of future technology cannot ignore the important social factor that is the moral compass (or lack of it) among its users and thus an intelligent technology should also be aware of the socio-cultural impact of ethics, what influences it and how it evolved with time. At the same time, whoever creates the technology, needs to take into account the ethical self of the user and be aware of the fact that adherents to different cultures, born in different times and thus also having different definitions of what is morally correct and not, have different demands towards technology.

It has not been yet measured to what degree the change in the way children are introduced to moral principles through the school curriculum is reflected in the broader mindset of the population and visible in associations with certain ethical categories and societal key concepts before and after major historical events, such as for instance the Second World War. In this study, we aimed at verifying how much does ethics, as taught in Japan in the course of the mandatory school curriculum before the Second World War and as taught right now has in common with the moral attitudes of contemporary Japanese and the way common people approach core moral questions (such as those concerning the sacredness of human life). We were also interested in finding out whether a big shift in the contents of lessons taught as part of the subject "Ethics" in schools would resonate within society by influencing what the larger masses think is morally acceptable or not. We tried to confirm this by selecting textbooks used in teaching ethics at Japanese school from between 1935 and 1937, and those used in Japanese junior high schools today (2019) and analyzed what emotional and moral associations were generated by the textbooks content. The analysis was performed with an automatic moral and emotional reasoning agent and based on the largest available text corpus in Japanese as well as on the resources of a Japanese digital library.

The remainder of this paper is structured as follows. We give a brief outline of Japanese Educational Reforms and their influence on teaching Ethics in Section 2, introduce the resources used in this study in Section 3, and describe the method used to analyze obtained data in Section 4, followed by the results of the experiment. Section 5 consists of a discussion and proposed directions for further research. Finally, the paper is concluded in Section 6.

## 2 Hundred Years of Japan's Educational System

Japan has a long history of compulsory schooling starting as early as 1886, fourteen years after the issuance of the 1872 Education Code, which was the globally first comprehensive plan for mass schooling.

Elementary School Regulations (*Shōgaku Kyōsoku*), issued in September 1873, became the impulse to the introduction of a new school subject called *shūshin* („teaching of good virtue").

In 1890, the Imperial Rescript on Education (*Kyōiku ni Kansuru Chokugo*) was signed to articulate government policy on the guiding principles of education in the Empire of Japan. The main objective of the rescript was the education of an "ideal imperial subject"[2], and it promoted core moral values such as piety, loyalty, friendship, benevolence, sincerity, building of prosperity, respect, courage, modesty, obedience, docility, conformity and submission of children to parents and teachers, nation, and the emperor. As of 1910, more than 98% of primary school-aged children attended four year compulsory primary school.

There were six original subjects taught in the 1880s:

- *shūshin*
- reading
- writing
- Japanese calligraphy
- mathematics
- physical education
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among which *shūshin* was considered the most important subject.

After Japan's defeat in the Second World War, the "Order of Suspension of Courses in Morals, Japanese History and Geography" by the American-led General Headquarters terminated *shūshin* in December 1945 as part of a transition from pre-war militarist policies to a new law. In 1947, following the end of the Second World War, a new law, the Fundamental Law of Education (FLE), the Fundamental Law of Education (FLE), for which the central philosophy was to confirm equal rights to education and to democratize the country. In terms of subjects taught at school, pre-war worship of the Emperor taught through *shūshin* was replaced with an alternative subject, *dōtoku* (道徳, roughly translatable as "Ethics"), with promoted core moral values including self-sacrifice for the good of others, conformity over individuality (derived from Confucianism) as well as self-awareness and the development of moral thinking [Duke, 2009].

"The General Course of Study" in 1951 showed the direction of moral education not as a teaching subject in school but as a subject interwoven through the whole school education.

However, "Report of the Curriculum Council: Establishment of Special Time of Moral Education," in 1958, pointed out that the whole-school approach did not produce effective education and suggested that a period for moral education should be established formally.

The present Basic Act on Education[3] mentions that education should "cultivate morality and ethics". MEXT[4] started to provide free supplemental learning materials called *Kokoro no Nōto* or "Notebook for the Heart", in 2002.

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[2] The 315-character document was read aloud at all important school events, and students were required to study and memorize the text. A translation of the edict can be obtained from: https://en.wikisource.org/wiki/Imperial_Rescript_on_Education

[3] A translation of the Basic Act on Education can be found here: https://www.mext.go.jp/en/policy/education/lawandplan/title01/detail01/1373798.html

[4] The Japanese Ministry of Education, Culture, Sports, Science and Technology.

In 2006, the Fundamental Law on Education was revised for the first time in 59 years. A major controversy resulted from its emphasis on "tradition," "discipline," "morality," and most of all, "patriotism" or "love of nation," all of which seem reminiscent of the 1890's Imperial Rescript on Education.

Further change in the way morality is taught in Japanese schools is ahead as the Central Council for Education, an advisory body for the education minister, has submitted a report calling for upgrading moral education to an official school subject[5] [Sakurai, 2016] and has submitted a report calling for upgrading moral education to an official school subject. The education ministry will soon start preparations to introduce it on a par with traditional subjects like Japanese, mathematics, social studies and science at elementary schools and junior high schools, beginning in fiscal 2018 and 2019.

In this work, we will analyze ethics as taught in Japan before WWII and today to verify how much the pre-WWII moral attitudes have in common with those of contemporary Japanese, to what degree what is taught as ethics in school laps over the general population's understanding of ethics and finally to try to assess whether a major reform of the guidelines for teaching the school subject of "ethics" at school after 1946 has changed the way common people approach core moral questions and what part of ethics, if any, was left missing after the education system reforms.

## 3 Applied Resources

### 3.1 Language resources to analyse

***Shūshin kyōjuroku*** (修身教授録, "Records from lectures in moral education") [Mori, 1989] is a record of the content of the seventy nine classes that Mori Nobuzō, a Japanese philosopher and educator, gave on the subject of *shūshin* between 1937 and 1939. Although the participants of the course were to become teachers, he conducted a course on general ethical matters, such as „the question of life and death" or „one only lives once," etc., amply enlarged through incorporating stories based on the life of role models for future educators.
The book is a long seller sold in over 150 000 copies was first published in 1989 and had over 50 new editions.

***Watashitachi no dōtoku*** (私たちの道徳, "Our morals") [MEXT, 2014] is a textbook for the subject "Ethics" taught in Japanese primary and junior high schools. MEXT started to provide free learning materials called *Kokoro no Nōto* or "Notebook for the Heart" (心のノート), in 2002. In 2014, MEXT decided on a major reform of the content of this textbook and printed it again under the title of *Watashitachi no dōtoku*. It is distributed for free to all Japanese primary and junior high schools. The book's edition used in this study is divided into four main chapters, covering topics such as „Living as a member of society" or „Supporting one another".

We used the two-abovementioned resources, one used in schools in Japan the 1930s and one still in use today as input to be analyzed with an automatic moral and emotional reasoning agent, in order to verify

[5] The education ministry introduced it on a par with traditional subjects like Japanese language, mathematics, social studies and science in elementary schools and junior high schools, beginning from April 2018.

to what degree what is taught as ethics in school overlaps with the general population's understanding of ethics.

### 3.2 Language resources for application in information processing tools

**Aozora Bunko** (青空文庫, literally the "Blue Sky Library", also known as the "Open Air Library") is a Japanese digital library that encompasses several thousands of works of Japanese-language fiction and non-fiction, including out-of-copyright books or works that the authors wish to make freely available.

*Aozora Bunko* was created on the Internet in 1997 to provide broadly available, free access to Japanese literary works whose copyrights had expired. Most of the texts provided are Japanese literature, and some translations from English literature. The resources are searchable by category, author, or title. The files can be downloaded in PDF format or simply viewed in HTML format.

In 2013, the Future of Books Fund (本の未来基金 *Hon no mirai kikin*) was established independently to assist funding and operations for *Aozora Bunko*. *Aozora Bunko* currently includes more than 15,100 works, a majority of which are novels, as of 5 January 2019.

**Yet Another Corpus of Internet Sentences (YACIS)** [Ptaszynski et al., 2012] is the largest Web based blog corpus available for Japanese language, collected automatically from the pages of Ameba blog service and containing 5.6 billion words within 350 million sentences.
We used YACIS first as the corpus of contemporary Japanese based upon which we checked for emotional and moral associations with ethical teachings from both the 1930s and today's Japan.

We then performed a second experiment, in which we compared the contents of the older textbook, *Shūshin kyōjuroku*, with a sample created of books published before 1946 present in the Aozora Bunko library to see to which degree what was taught as part of ethics classes in Japanese schools pre-Second World War overlapped with the moral compass of larger society.

### 3.3 Information processing tools

**Moral Reasoning Agent (MRA)**, first proposed by [Rzepka and Araki, 2005] and further developed by [Komuda et al., 2010]. The moral consequence retrieval agent was based on the idea of Wisdom of Crowd. In particular, it uses a Web-mining technique to gather consequences of actions applying causality relations, to extract from the Web emotional and ethical consequences of actions found in the input[6]. The agent was previously tested on over 100 ethically significant real world statements, such as "killing a man", "stealing money", "bribing someone", "helping people" or "saving environment".

The agent takes a sentence as an input and in a specified corpus (such as YACIS or Aozora bunko, as in our experiment) searches for emotion- and morality-related concepts associating with the sentence

[6] The agent takes a sentence as an input and in a specified corpus (such as YACIS) searches for emotion- and morality-related concepts associating with the sentence contents. The technique is composed of five steps: a) extraction of input phrases; b) modification of the phrases with causality morphemes; c) searching for the modified phrase in the specified corpus; d) matching to the predetermined lexicons (containing moral and emotional concepts) and extraction of emotion associations; e) ranking creation and output. The higher hit-rate an expression had in the Web search, the stronger was the association of the original phrase to the emotion and moral types. For example, a phrase "thank you" is likely to associate with gratitude, relief and joy and "saving a person" is more likely to associate with such moral associations as "deserve praise", or "thankfulness".

contents. The technique is composed of five steps: a) extraction of input phrases; b) modification of the phrases with causality morphemes; c) searching for the modified phrase in the specified corpus; d) matching to the predetermined lexicons[7] (containing moral and emotional concepts) and extraction of emotion associations; e) ranking creation and output. A model of the functioning of the Moral Reasoning Agent is demonstrated in Figure 1. The higher hit-rate an expression had in the Web search, the stronger was the association of the original phrase to the emotion and moral types. For example, the phrase "saving someone" is likely to associate with gratitude, relief and joy and match with ethical consequences such as "be praised by people" while "killing a person" is more likely to be linked with such ethical consequences as "going to jail", or "condemn".

Through running two datasets created on the basis of the two schoolbooks with the Moral Reasoning Agent, we performed a complete analysis of moral and emotional associations with the ethical teachings deemed useful in the 1930s and nowadays, to see how they stand up to a broader societal standard.

The whole process is explained in more detail in the following section.

## 4 Quantitative Analysis

### 4.1 Initial Data

As a first step in preparing our data sample, we performed an OCR of *Shūshin kyōjuroku* and *Watashitachi no dōtoku* and performed sentence segmentation of both texts. Due to the size of the initial sample, based on both texts, which contained 1,665 sentences, we chose to only compare chapters referring to the same areas of morality – in this case the sacredness of human life, approaches to human life as a valuable gift, a meaningful life and similar[8].

#### 4.1.1 Creating Evaluation Dataset from *Shūshin kyōjuroku*

Upon selecting the chapters in question, we created a list of grammatical endings that might imply that the sentence contains an ethical warning or positive evaluation of a given action. The endings contained grammatical forms such as *ga yoi deshō* ("It would be a good thing to…") or *toiu wake desu* ("It is so, that…").

There were 172 initial results, split into „negative sentences" (e.g. „It is not…", „You shouldn't", etc.), „positive short" („Let's…") and „positive long" („It would be a good thing to…") sentences. We then manually proceeded to remove sentences which brought no ethical meaning to them.

This left us with a sample of 77 sentences which we then evaluated with the Moral Reasoning Agent.

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[7] The lexicons were based on ML-Ask. The emotional lexicon contains 2100 items (words and phrases) describing emotional states, divided into 10 emotions classes. The ethical consequence lexicon was created through substituting the ten emotion types into five pairs of word groups representing Kohlberg's stages of moral development (based on Lawrence Kohlberg's theory of human moral development, in which he assumed successive changes in aspects by which we consider an action good or wrong). The items in the lexicon were distributed as follows: Praises (18) / Reprimands (33); Awards (25) / Penalties (15); Society approval (8) / Society disapproval (8); Legal (8) / Illegal (8); Forgivable (6) / Unforgivable.

[8] Sub-chapters: 2 (*Ningen to umarete*, , "To be born as a human"), 6 (*Jinsei no shūshi*, "The beginning and end of one's life") , 37 (*Shisei no mondai*, "Questions related to life and death") from part one and 3 (*Jinsei nido nashi*, "You only live once"), 4 (*Seimei no aiseki*, "The caress of life"), 14 (*Jinsei no fukasa*, "The depth of life") and 29 (*Jinsei ha myōmi shinshin*, "Life is mysterious") from part two of *Shūshin kyōjuroku* and chapters 3-1 (*Kakegae no nai jita no seimei wo sonchō shite*, "Respecting the irreplacable life of yourself and others"), 3-2 (*Utsukushii mono he no kandō to ikei no nen wo*, "Let yourself be impressed and admire beautiful things") and 3-3 (*Ningen no tsuyosa ya kedakasa wo shinji ikiru*, "Live by believing in human strenght and nobility") from chapter 3 of *Watashitachi no dōtoku*, entitled *Seimei wo kangaeru*, "Thinking about life".

### 4.1.2 Creating Evaluation Dataset from *Watashitachi no dōtoku*

Since there is much less text in this textbook (due to a large number of illustrations as well as putting an emphasis on interaction and thus containing exercises to be filled in by the student), instead of choosing grammatical constructions prone to containing ethics-related statements, we kept all sentences containing an ethical evaluation, which left us with a total of 52 sentences to run through the Moral Reasoning Agent.

The grammatical information contained in those sentences also differed from *Shūshin kyōjuroku* in the sense that much less was in the form of ready-made teachings delivered by a higher authority and much more often in the form of open questions ("What do you think about…" or "Have you ever given reflection to the question of…").

## 4.2 Challenges in Comparison of Two Textbooks

*Shūshin kyōjuroku* is a written report of a teacher talking and giving life lessons, very rarely a Q&A. Grammatically, it contains a number of different structures, with a lot of emphasis on „what one ought to do", „The right thing to do" – the duties of a human being, as well as one's relations with those above and below in a hierarchical social structure. A part of the book is devoted to studying the lives of significant individuals influencing Japanese ethical thought, such as Shinran[9] or Johann Heinrich Pestalozzi[10]. Since it is a *compte-rendu* (formal report) of a series of lectures intended specifically for pedagogy students, a large part of the material is devoted to questions related to education and upbringing.

*Watashitachi no dōtoku* sometimes covers a wholly different range of topics, such as "The benefits and drawbacks of living in an information society" or "Contributing to the world as a real international person while being aware of being Japanese," which are absent from the content of *Shūshin kyōjuroku*. The narration is mostly in the style of open questions („What do you think would be the right thing to do?") as well as exercises intended to engage the reader („Imagine your behavior in this situation", „Write down your thoughts upon reading this chapter").

The process of creating two datasets from *Shūshin kyōjuroku* and *Watashitachi no dōtoku* to be analyzed using the Moral Reasoning Agent is shown in Figure 2.

## 5. Results and Discussion

## 5.1 First Experiment Results

Upon analyzing the two datasets with the Moral Reasoning Agent, we were left with 22 sentences for which the MRA found enough sample sentences in the corpus to enable it to run a complete analysis[11] of

[9] Shinran (親鸞, 1173-1263) was a Japanese Buddhist monk, the founder of the Jōdo Shinshū sect in Japan.

[10] Johann Heinrich Pestalozzi (1746 –1827) was a Swiss pedagogue and educational reformer who founded several educational institutions both in German- and French-speaking regions of Switzerland in order to overcome illiteracy.

[11] Containing, apart from an overall evaluation of moral and emotional consequences, a detailed description of moral («worth praise», «correct action») / immoral («illegal act», «worth reprimanding») associations with the sample term.

moral and emotional associations (among the results, 8 were based on the sentence sample from *Shūshin kyōjuroku* and the remaining 14 on *Watashitachi no dōtoku*).
A detailed analysis for a sample sentence *hito no inochi wo sukuu* ("to save people's lives") is presented in Table 1.

More striking results of the first experiment were summed up in Table 2, where upon presenting the original sentences, ethically relevant statements extracted from those sentences and sample sentences containing the same ethically relevant statements found in YACIS, for further clarity we emphasized examples where there were surprising twists in terms of the percentage of positive and negative emotional consequences of the actions.

The largest difference in approach in the results of the first experiment is visible when taking into account the emotional value attributed to certain behaviors - with the sample from more recent material giving far more negative or ambivalent responses – which would indicate a high level of consciousness of the fact that behaving in the most moral way, beneficial for the broader society is not always pleasant or beneficial to the individual undertaking the action.

### 5.2 Second experiment results

Of course in order to analyze the mentality of the average Japanese living in the 1930s it would be ideal to compare the sample from *Shūshin kyōjuroku* with a corpus of sentences from the same time period (namely the 1930s); however, at the time this paper was written the authors were not aware of the existence of such a corpus. We decided to settle for the next best thing-to replace the corpus used by the Moral Reasoning Agent and instead of YACIS use texts from Aozora Bunko[12], a Japanese digital library, published before 1946 and thus influenced by ethical ideas from the 1930s and before. As a first task we segregated the corpus- since it is not classified according to time periods and there is no research engine that would allow for a robust selection of texts, we chose to retain only works for which the date of first publication was given. We also removed all translations of foreign works (all items with foreign title given in the *gendai*- 原題, original title- category, of which there were 134 items in total). Finally, we removed items coming from before the Meiji era (1868; 4 items, 2 distinct works). This preliminary data cleaning left us with a total of 1873 texts published before 1946. We thus enabled the program to simulate a pre- Second World War moral outlook[13].

This time, out of 58 terms used in the analysis, the Moral Reasoning Agent found enough sample sentences in the corpus to enable it to run a complete analysis of moral and emotional associations for a total of 45 terms. The most interesting examples of the results of the second experiment were summed up in Table 3.

### 5.3 Discussion

[12] https://www.aozora.gr.jp/

[13] While of course being aware that this solution has certain limitations, as book authors form only one group of socjety, genrally better educated and having more liberal opinions than an average member of society. There is also the question of whether the contents of a work of litterature represent the opinions of the author.

In the results of both the first and the second experiment one can perceive a discrepancy between behaviors beneficial to the individual engaged into a certain action and to broadly perceived society- in terms such as 苦労 (*kurō*, suffering) or 正直 (*shōjiki*, honest).

This sharp distinction between understanding what is beneficial to society as a whole and what is beneficial for the person performing a certain positive action is in line with Charles Leslie's Stevenson's and Alfred Jules Ayer's emotivism (the hurrah/boo theory) [Stevenson, 1944] – a meta-ethical view that claims that ethical sentences do not express propositions but emotional attitudes and the presence of an ethical symbol in a proposition adds nothing to its factual content [Ayer, 1936]. As such, while the respondent to certain ethical claims understands their positive value objectively, the overall positivity of the response is reduced due to the consciousness of its plausible negative emotional impact on the individual in question.

In opposition to the view that conscience is the "inner light", Jeremy Taylor, Christian thinker from 17th century, famously states, "conscience is, in most men, an anticipation of the opinions of others". William R. Alger similarly claims, "Public opinion is a second conscience". This directs us to the assumption that conscience can be perceived as an approximate result of the opinions of other people, and thus openly assessing certain actions as morally correct or being "the right thing to do", all the while maintaining a "private ethic" of sorts, is a natural copying mechanism of humans, notwithstanding external circumstances.

This theory proves interesting in the case of more controversial terms such as 国土 (*kokudo*, the territory of one's country, motherland) or 国民 (*kokumin*, nation- the complete results for the two terms are shown in Table 4 and 5), while giving an inconclusive assessment in terms of emotional consequences, are judged positively in terms of benefit to society but above all present in both the textbook for teaching ethics and the litterature of the 1930 and before which served as our corpus in this experiment are included in the textbook for teaching ethics used in middle school nowadays but at the same time completely absent from a corpus of modern Japanese (and thus one might say from the conscience of the broader population nowadays)[14].

Do those results indicate that while subject to a school curriculum centered around values such as nation, serving others, devotion to the motherland, the Japanese chose to superficially believe or attribute value to what they were taught was morally proper in ethics lessons while privately maintaining a more inconclusive approach to several ethical problems? Did they completely lose interest in ethical matters related to strongly patriotic ideals after 1946?

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[14] *Watashitachi no dōtoku* contains the following four subchapters to its Chapter 4 (社会に生きる一員として、*Shakai ni ikiru ichiin to shite*, Living as one of the members of a society): ふるさとの発展のために(*Furusato no hatten no tame ni*, For the developement of one's hometown), 国を愛し、伝統の継承と文化の創造を (*Kuni wo aishi, dentō no keishō to bunka no sōzō wo*, Love the country, inherit the tradition and create culture), 日本人としての自覚を持って真の国際人として世界に貢献したい(*Nihonjin to shite no jikaku wo motte ma no kokusaijin to shite sekai ni kōken shitai*, Contributing to the world as a truly international person while maintaining the awareness of being Japanese*)*, 日本人の自覚を持ち世界に貢献する(*Nihonjin no jikaku wo mochi sekai ni kōken suru*, Contributing to the world while maintaining the awareness of being Japanese).

We might state that the reform of education undertaken in 1946 had its biggest effects in this aspect- terms broadly associated with nationalism, while still subject of focus of school- taught ethics, seem to leave the broader society indifferent.

While in terms of core moral values there is little difference to be observed between the moral evaluation of an action between sentences from both *Shūshin kyōjuroku* and *Watashitachi no dōtoku* and YACIS – meaning what was taught as morally proper in the textbooks juxtaposes with what the general public finds morally proper, which would suggest that in spite of a huge shift in terms of which ethical values were promoted through school education, the understanding of morality, a „moral compass" of the Japanese people remained largely unchanged- queries for phrases evolving around the sanctity or value of life always returned positive results when it came to assessing the impact on society (for instance in the case of the query "give yourself to people", 人のために尽くす, *hito no tame ni tsukusu*, for which a detailed analysis is shown in Table 6, both a sample based on YACIS- the way Japanese people assess this action now- and Aozora Bunko- the way they thought of sacrifice for the sake of others in the 1930s- showed positive social consequences). This both heartening (no amount of mistaken moral education motivated by short-term political agenda seems to make people lose their integral sense of what is right and wrong) and at the same time disheartening (a huge rewriting of the basic principles governing mandatory education in the area of ethics seems to have, again, a rather limited impact on the core morals of a society) seems to confirm the definition of ethics refer to rules provided by an external source, e.g., codes of conduct in workplaces or principles in religions and morals understood as an individual's own principles regarding right and wrong.

This would provide some insight as to how the Japanese people, fed lessons on the vertical construction of society together with militaristic and nationalistic content all through the 1930s and during the Second World War right until the country's surrender in August 1945 would then shake off the of their entire education and acclaim the promulgation not even two years later, in May 1947, of a pacifist constitution, proclaiming the equality of all men and renouncing the nation's right to belligerency (Dower, 1999).

Assuming that the real impact of ethics understood as a set of rules taught throughout the school curriculum seems to have only a limited influence on the morality of an individual, and in light of the reforms undertaken recently by the Japanese government, how should ethics lessons be organized? Is it possible to condition a child's moral compass through ethics classes? And finally, is it possible at all, as the Japanese Ministry of Education seems to imply, to evaluate and grade a student's ethical outlook in the same way you would grade homework in mathematics?

### 5.4 Future directions

As a next step to deepen the analysis of this research, our plan is to perform the same process on two more samples, one related to work matters (relationships within one's workplace as well as with subordinates and superiors) and one related to patriotism (feelings towards one's home country), as those two areas of ethics are covered by both textbooks and we can expect a shift occurring together with Japan's period of post-war with the professional landscape dominated by big companies and the shift of the country's politics from militarism towards pacifism.

Furthermore, we would like to find out whether, as the results of our second experiment would suggest, the Japanese really lost interest in ethical matters related to concepts such as the motherland or the nation, through performing a survey analysing their approach, assessment and means to express patriotism.

## 6. Conclusions

The objective of this paper was to demonstrate the challenges facing educators trying to implement ethics as part of the school curriculum, on the example of the contents of textbooks for the subject of „Ethics" in school in pre- and post-war Japan. We were particularly interested in finding out whether changes in ethical education influenced moral core values in humans throughout the century, how much pre-WWII moral attitudes have in common with those of contemporary Japanese, and whether a major reform of the guidelines for teaching the school subject "Ethics" at school after 1946 has changed the way common people approach core moral questions (such as those concerning the sacredness of human life).

For this purpose, we selected textbooks used in teaching ethics at school from between 1935 and 1937, and those used in junior high schools today (2020) and analyzed what emotional and moral associations such contents generated.

We found out that past and contemporary Japanese share a similar moral outlook on certain basic moral concepts, although there is a discrepancy in the way in which they evaluate those concepts in terms of benefit to society and emotional consequences, which would indicate awareness in the society of the dissimilarity between the categories of ethics understood rules provided by an external source and morals understood as an individual's own principles regarding proper and improper conduct.

Some phrases containing an ethical evaluation associated with a nationalistic trend in education, assessed positively in the pre- Second World War sample have also disappeared from the scope of post-war schoolbooks. The findings of this study support suggestions proposed by others that the development of personal AI systems requires supplementation with moral reasoning, through indicating that while ethics are subject to evolution, due for instance to historical events or political agendas, there exists the need to balance them with morals, which at its core remain largely uninfluenced by external factors. If (and when) personal AI systems become obtain a singularity (a self-conscious state- Kurzweil (2005)), in order for this technology to respond to the private needs of a diversified range of users it must include an insight into culture- related and morality- related, pluralistic demands and needs of its potential users.

## Appendix

| | | | | | | |
|---|---|---|---|---|---|---|
| Phrase | *hito no inochi wo sukuu* (“to save people's lives”) | | | | | |
| Original sentence in *Watashitachi no dōtoku* | 人の命を救い、 人々の苦しみを 和らげる以外に考えることは 何もない。 | | | | | |
| Translation | There is nothing to think of except saving people's lives and alleviating their suffering. | | | | | |
| Sample sentence in YACIS | 医療の現場ってとても想像できなくて（特に救命なんてね）、人の命を救うプレッシャーをずっと受け続ける現場なんて、やりたいって思う人なんてめったにいないですよね〜。 | | | | | |
| Translation | I can hardly imagine what it feels like to be working in ER (especially saving lives), and rarely do people want to do a job where you are under pressure to save human lives, right? | | | | | |
| Frequency in the corpus | 294 | | | | | |
| Social consequences (positive/negative) | 100% / 0% | | | Emotional consequences (positive/negative) | 50% / 49% | |
| Moral categories | Correct action | 23% of the results | | Immoral categories | Illegal act | n/a |
| | Worth encouraging | 14% | | | Unacceptable conduct | n/a |
| | Worth praise | 4% | | | Worth reprimanding | n/a |

**Table 1 -** A detailed analysis for a sample sentence *hito no inochi wo sukuu* (“to save people’s lives”)

| Sample phrase (W – *Watashitachi no dōtoku*, S – *Shūshin kyōjuroku*) | % of positive / negative consequences | | Sample sentence from YACIS |
|---|---|---|---|
| | Social | Emotional | |
| 感謝の気持ちを持つ (“feel gratitude”) – S | 100% / 0% | 57% / 42% | 感謝の気持ちを持つことは、習慣のようなところがあり、なれないと、違和感があるかもしれません (“Being grateful is like a habit, and if you don't develop it, you may feel uncomfortable”) |
| 一人で悩まない (“not suffer on your own”) – S | 50% / 50% | 51% / 48% | 一人で悩まないで誰か身近な人にどんどん話そう。ある程度を超えるとさすがにうざいけど吐き出していい。(“Don't worry on your own and talk to someone close to you. At some point, though it might be embarassing, you just have to get it out of your system”) |
| 思いやりを持つ (“be empathic”) – S | 85% / 14% | 13% / 86% | 挫折した時も、痛みが分かる優しさや思いやりを持つために生まれてきたんだよって支えてくれた母。(“My mother supported me when I was frustrated and explained that I was born with kindness and compassion to understand the pain”) |
| 人を励ます (“encourage people”) – W | 66% / 33% | 55% / 44% | 気持ちも落ち込みがちです。 こんなオイラは他の人を励ます資格あるんだろうか。(“I also tend to be depressed. Makes me wonder whether someone like me is qualified to encourage others?”) |
| 人はいつか死ぬ (“we will all die someday”) – S | 33% / 66% | 38% / 61% | 生と死、仕事に対する誇り、自分を理解してくれる人生のパートナー、色々な人の人生があること、人はいつか死ぬこと。(“Life and death, pride in one's work, a partner in life who understands me, the life of various people, and the fact we will all eventually die”) |
| 人の命を救う(“to save people's lives”) – W | 100% / 0% | 50% / 49% | 医療の現場ってとても想像できなくて（特に救命なんてね）、人の命を救うプレッシャーをずっと受け続ける現場なんて、やりたいって思う人なんてめったにいないですよね～。(“I can hardly imagine what it feels like to be working in ER (especially saving lives), and rarely do people want to do a job where you are under pressure to save human lives, right?”) |
| 人間は弱い (“humans are weak”) – W | 33% / 66% | 33% / 66% | 元々人間は弱いでも世のため人のために生き始めたときどんどんどんどん強くなれるどんどんどんどん美しくなれる人間不思議なるもの (“Originally, human beings are weak, but when they start living for others, they can become stronger and stronger, and become more and more beautiful, which is remarkable”) |
| 人を歓迎する (“welcome people”) – S | 0% / 100% | 57% / 42% | 今日はゲストが多く、２人で１人を歓迎するという感じだったので、負担が大きく大変だったのですが、お腹いっぱい食べることができ、すごくしあわせでした。(“There were so many guests today and it seemed like two people would welcome one, so the burden was big and it was hard, but I was able to eat a lot which made me very happy”) |

**Table 2 -** Percentage of positive and negative emotional consequences of ethically meaningful phrases in Experiment 1

| Sample phrase from *Shūshin kyōjuroku* | % of positive / negative consequences | | Sample sentence from Aozora Bunko |
|---|---|---|---|
| | Social | Emotional | |
| 人生 („life, human existence") | 62%/37% | 47%/52% | 今日から新しい自分の人生が始まるのだ、そういううちから強い感情が胸いっぱいに溢れて、家のなかにじっとしていられない気持だった。(From today, my new life will begin!- and as I said this, my chest was full with strong emotions and I couldn't stay still in my house). |
| 人間 („humanity, men") | 48%/51% | 36%/63% | ただ自分だけの子にするのではなく、御国の役にたつ人間、りっぱに御奉公のできる武士にしたい。（I do not want it to be only my child, but to become a human being who can serve the country, a samurai who can perform public duties splendidly）. |
| 仕事 („occupation, life's work") | 65%/34% | 43%/56% | そのあいだ縫い針洗濯の手仕事をしても、どんな事をしてでもきっと辛抱しとおしてみせます （In the meantime, no matter if you work washing or sewing I will be patient with you). |
| 苦労 („suffering, effort") | 51%/48% | 38%/61% | 短い人の一生をそんな取り越し苦労をして更に短くするのは、天命に背く罪とも言うべきじゃ。(It can be said that it is a sin against the natural order of things to make the short life of a person even shorter by making her endure such suffering). |
| 性欲 („libido, sexual desire") | 100%/0% | 50%/50% | 小説家とか詩人とかいう人間には、性欲の上には異常があるかも知れない。(People such as novelists and poets may well have an abnormal libido). |
| 誠 ("truth, reality") | 51%/48% | 35%/64% | 夫が決して人の情を解しないやうな人物でないばかりか、日本人としての意気と誠を十分にもつた男だと信じれば信じるほど、こんな些細なことに偏窟な自尊心をみせる気が知れない。(Not only is my husband one to understand other people's feelings, but what's more I believe that he is a man who has enough morale and sincerity as a Japanese person, and I find it hard to believe that such a trivial thing would lead him to demonstrate such mistakenly understood self-esteem). |
| 礼 („politeness, respectfulness") | 68%/31% | 41%/58% | とにかくに等しく恩のあるものならば、一方より礼を言いて一方より礼を言わざるの理はなかるべし (Anyway, if if you are equally endebted to two people, it should not be that you thank one without thanking the other). |
| 正直 („honesty, telling things as they are") | 66%/33% | 38%/61% | 正直にいって、そのとき志保は初めて妹に妬みを感じた。 (To be honest, at that time, Shiho felt envious of her sister for the first time). |
| 生命 („life, creation") | 53%/46% | 44%/55% | この法によれば、平民の生命はわが生命にあらずして借り物に異ならず。(According to this law, the life of a commoner is not equal in value to my life and is no different from an object you would lend). |
| 理想 („ideal, perfection") | 63%/36% | 41%/58% | それだのに、女はおれを高邁な理想主義者だと思つてゐるらしく、なかなか誘惑してくれない。(Women seem to think that I am a true idealist, and do not try to seduce me). |

**Table 3 -** Remarkable results of Experiment 2

| Phrase | *kokudo* (“motherland”) | | | | | |
|---|---|---|---|---|---|---|
| Original sentence in *Shūshin kyōjuroku* | われはひとり人間としてこの地上に生まれたばかりでなく、この日本の国土に生まれたということは、さらに大きな喜びでなくてはならぬと思うのです。 | | | | | |
| Translation | Not only born on this earth as human beings, being born in the land of Japan specifically must be a great joy. | | | | | |
| Sample sentence in *Aozora Bunko* | どうすれば日本の国土に相応し、風景と調和し、無事の日にはこころよい住心地と、たのしい安全感とをあたえるような住宅の群れを作りあげて、いよいよわたしたちの愛惜の念を、深くかつ切なるものにし得るかを考えなければならぬ。 | | | | | |
| Translation | How can we create housing suitable for the land of Japan, in harmony with the scenery, and providing a comfortable living environment and a pleasant sense of safety on a plain, uneventful day? I have to think deeply about how I can transform my regrets into something deep and meaningful. | | | | | |
| Frequency in the corpus | 205 times | | | | | |
| Social consequences (positive/negative) | 58% /33% | | | Emotional consequences (positive/negative) | 52% /47% | |
| Moral categories | Correct action | 14% of the results | | Immoral categories | Illegal act | 4% |
| | Worth encouraging | 11% | | | Unacceptable conduct | n/a |
| | Worth praise | n/a | | | Worth reprimanding | 17 % |

**Table 4 -** A detailed analysis for a sample phrase, *kokudo* (“motherland”)

| Phrase | *kokumin* (“nation, one’s people”) | | | | | |
|---|---|---|---|---|---|---|
| Original sentence in *Shūshin kyōjuroku* | 諸君らが今日たどりつつある道は、諸君らの自覚の浅深いかんにかかわらず、とにかく生涯を、国民教育のために生きる道であります。 | | | | | |
| Translation | The road you are taking today is a lifelong road, regardless of your shallow awareness of it. It is a way to live for national education. | | | | | |
| Sample sentence in *Aozora Bunko* | われわれが死を決するところはそれとは違う、大義を顕彰するということはわれわれ自身の問題ではなく、この国民ぜんぶの系体に関するのだ | | | | | |
| Translation | When we decide to die is different, honoring the cause is not our own problem, but a question regarding the whole of this nation. | | | | | |
| Frequency in the corpus | 2500 | | | | | |
| Social consequences (positive/negative) | 61% /38% | | | Emotional consequences (positive/negative) | 44% / 55% | |
| Moral categories | Correct action | 9% of the results | | Immoral categories | Illegal act | 6% |
| | Worth encouraging | 18% | | | Unacceptable conduct | 2% |
| | Worth praise | n/a | | | Worth reprimanding | 7% |

**Table 5 -** A detailed analysis for a sample phrase, *kokumin* (“nation, one’s people”)

| Term | | 人のために尽くす [*hito no tame ni tsukusu,* "to serve others/ the people"'] |
|---|---|---|
| Original sentence in *Watashitachi no dōtoku* | | 我」に引掛っているとは、常に自分の利害を中心にして、人のために尽くすということの分からない人間だ- *ga ni hikkakatteiru to ha, tsune ni jibun no rigai wo chūshin ni shite, hito no tame ni tsukusu toiu koto no wakaranai ningen da*, „Someone who is an egoist is a person who always focuses on his own interests and doesn't understand that one should serve others" |
| Sample sentence in YACIS | | 悲しみと苦痛は、やがて"人のために尽くす心"という美しい花を咲かせる土壌だ, *kanashimi to kutsū ha yagate hito no tame ni tsukusu kokoro toiu utsukushii hana wo sakaseru dojō da*, „Sorrow and pain are the soil that will eventually bloom with beautiful flowers called "serving other people" |
| Frequency in the corpus | | 213 times |
| Social consequences (positive/negative) | | 100%/0% |
| Emotional consequences (positive/negative) | | 47% /52% |
| Moral categories | Correct action | 7% of the results |
| | Worth encouraging | n/a |
| | Worth praise | n/a |
| Immoral categories | Illegal act | n/a |
| | Unacceptable conduct | n/a |
| | Worth reprimanding | n/a |
| Original sentence in *Shūshin kyōjuroku* | | すなわち諸君たちが教育を通して国家社会に尽くす,*sunawachi shokun-tachi ga kyōiku o tōshite kokka shakai ni tsukusu,*"In other words you young people will contribute to the nation through education". |
| Sample sentence in Aozora Bunko | | 父子二人が身を捧げ、人の為にも尽くす, *fushi futari ga mi o sasage, hito no tame ni mo tsukusu, "*Both father and son will devote themselves and serve others". |
| Frequency in the corpus | | 71 times |
| Social consequences (positive/negative) | | 66%/33% |
| Emotional consequences (positive/negative) | | 19% 81/% |
| Moral categories | Correct action | n/a |
| | Worth encouraging | 16 % of the results |
| | Worth praise | n/a |
| Immoral categories | Illegal act | n/a |
| | Unacceptable conduct | n/a |
| | Worth reprimanding | 5 % of the results |

**Table 6 –** A lack of major shift in core morals of Japanese people demonstrated on the example of the phrase "give yourself to people", 人のために尽くす, *hito no tame ni tsukusu*,

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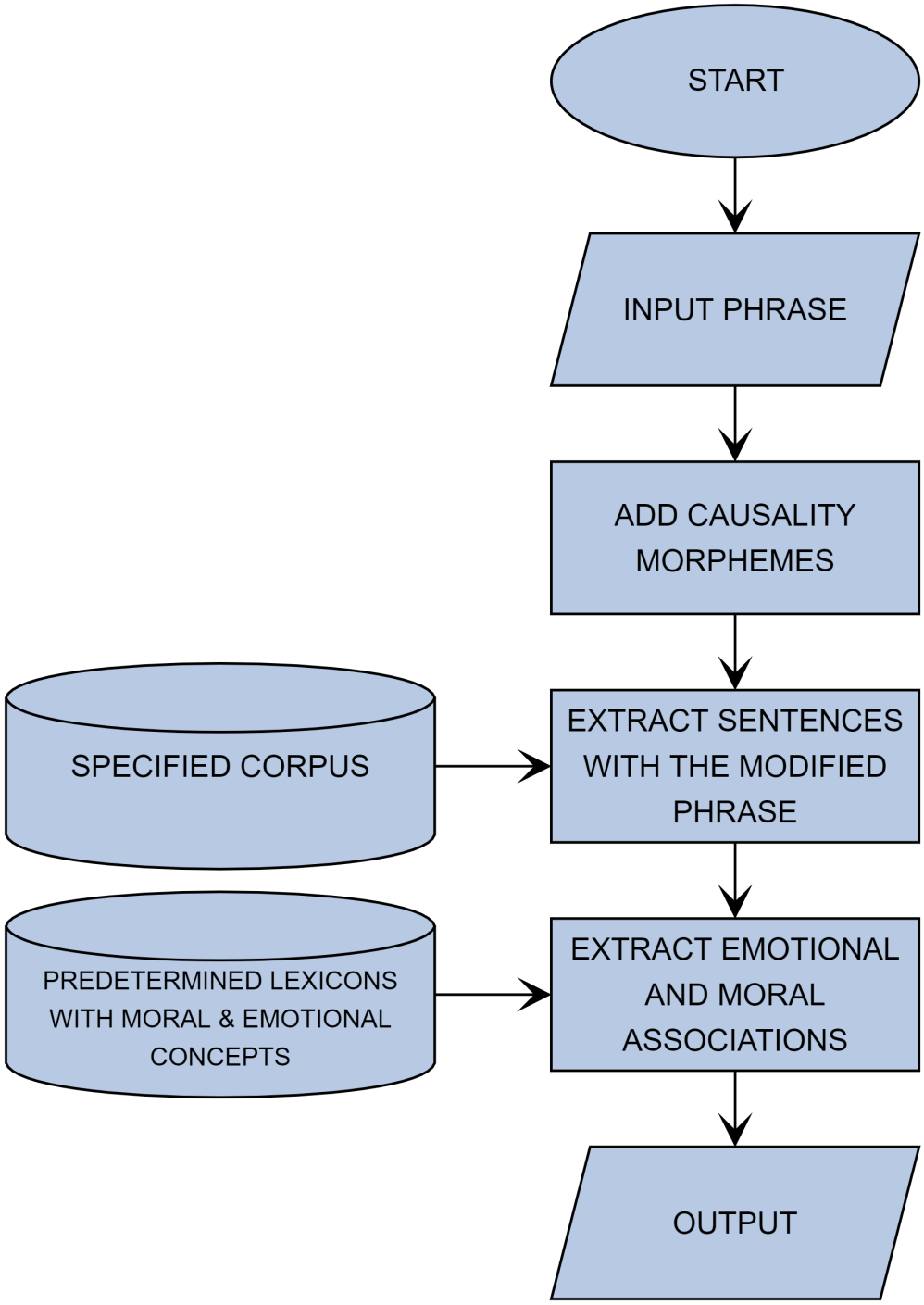



**Figure 1 -** Operation pattern of the Moral Reasoning Agent.

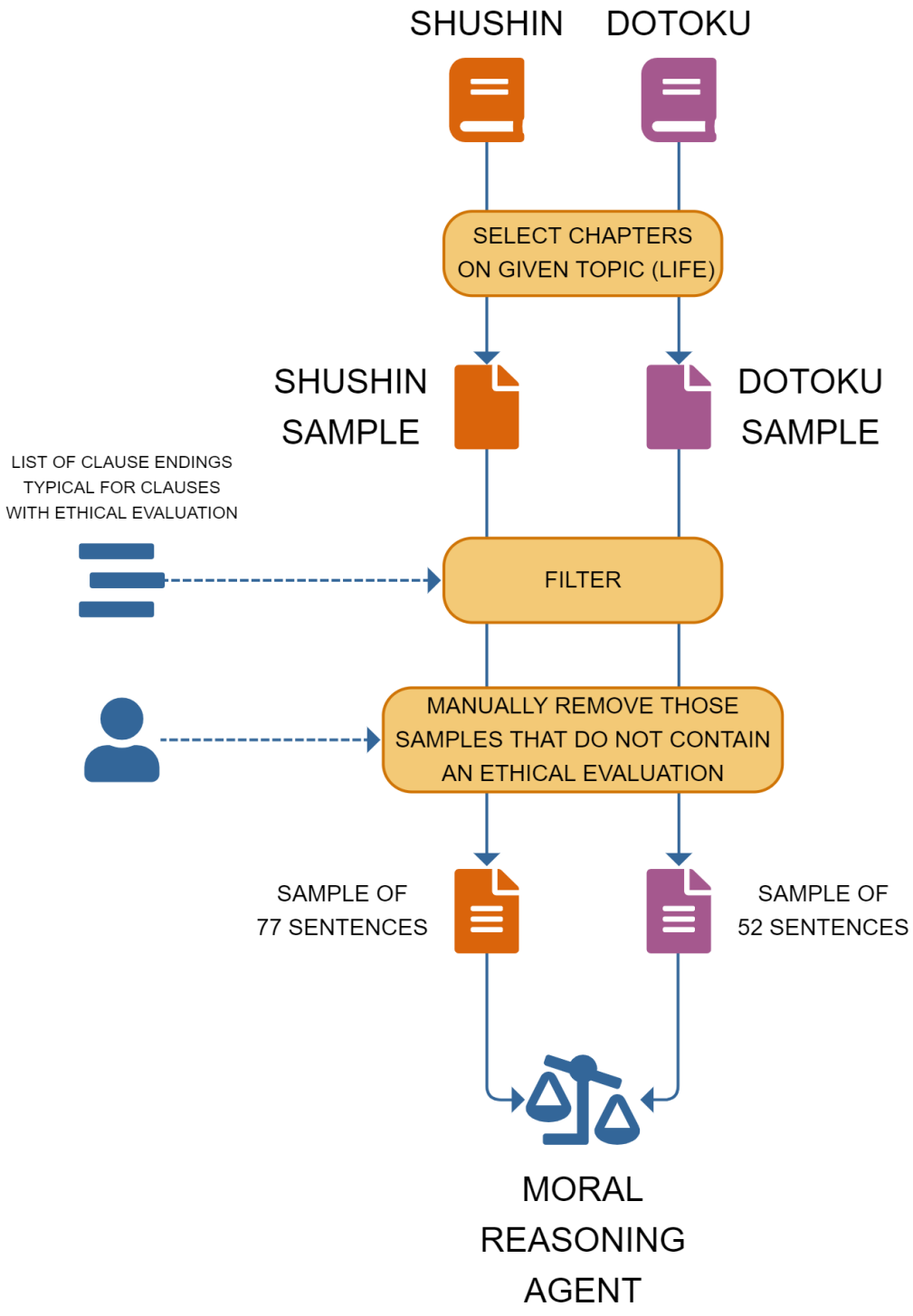



**Figure 2 -** Creating two datasets from *Shūshin kyōjuroku* and *Watashitachi no dōtoku* to be analyzed using the Moral Reasoning Agent